Dual Spatial-Temporal Shapley Attribution for Explainable Anomaly Detection in Dynamic Social Graphs

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***Abstract*—Transformer-based detectors such as TADDY find anomalous edges in dynamic social graphs with high accuracy, but they give no reason for an alarm: the analyst gets a score and nothing more. We present a post-hoc framework that wraps a trained, frozen TADDY model and returns two attribution vectors for each flagged edge: a *spatial* vector that says which neighbour roles drove the score, and a *temporal* vector that says which snapshot carried the signal. The framework adapts the GraphSVX Shapley surrogate to TADDY's windowed tokens through an embedding-space perturbation that replaces masked tokens with the mean of the active ones, instead of the misleading index-space masking. On three dynamic social graph benchmarks, detection is unchanged (ΔAUC $= 0$). In a controlled comparison on the same trained model, the framework gives the highest Comprehensiveness of all baselines on every dataset (Random, Attention, Gradient, Gradient$\times$Input; paired Wilcoxon $p < 10^{-10}$) and the highest Fidelity on UCI-Message, the dataset where Fidelity is not saturated. An ablation shows that embedding-space masking is the component that matters: index-space masking cuts Comprehensiveness by 41%. An expanded evaluation over $2{,}621$ edges (true positives, false positives and low-confidence detections) and a worked example show how an analyst reads the two vectors. One explanation costs about 0.18 s on a CPU core.**

***Index Terms*—Social network analysis, dynamic graphs, anomaly detection, explainability, Shapley values, temporal attribution, graph transformers.**

I. Introduction

Social networks are never static. Messaging graphs, trust and rating communities and platform interaction graphs change all the time as new edges appear. Finding anomalous behaviour in these graphs (coordinated inauthentic activity, abusive accounts, fraudulent ratings) is urgent in practice and hard in theory [1].

Deep detectors have raised accuracy a lot. NetWalk [2], AddGraph [3], StrGNN [4] and TADDY [5] use graph neural networks and Transformers to capture spatial and temporal patterns. TADDY stands out: it turns each candidate edge into a short sequence of tokens (neighbour rank, hop distance and time step) and feeds it to a BERT-style Transformer encoder. Benchmarks that re-run these methods under one protocol place it among the strongest detectors [6].

None of these detectors can explain itself. When edge (u, v) is flagged at snapshot t, TADDY returns a score. It does not say which of the surrounding nodes looked suspicious, or whether the anomaly appeared now or was building over earlier snapshots. Trust-and-safety teams cannot act on alerts they cannot understand, and regulated settings require automated decisions to be justified.

For TADDY, explanation quality metrics are not low: they are *undefined*, because there is no attribution to evaluate. This paper fills that gap. Our framework wraps a frozen TADDY model and, for each flagged edge, answers two questions at once: *was the alarm driven by a suspicious counterparty, or by a suspicious moment in the interaction history?*

Contributions.

1) The first post-hoc explainability framework for TADDY-style detectors, producing **dual spatial and temporal attribution vectors** for each flagged edge, with zero change to detection.
2) A **graph-aware embedding-space perturbation** that replaces excluded tokens with the mean embedding of the active coalition. An ablation shows it is the essential component of the framework.
3) A **controlled comparison** against four baselines on the same trained model, with bootstrap confidence intervals and paired Wilcoxon tests: the framework has the highest Comprehensiveness on all three datasets.
4) An **expanded evaluation over $2{,}621$ edges** (all true positives, false positives and low-confidence detections), a worked example, a top-k sensitivity study and a cost analysis.

II. Related Work

A. Anomaly Detection in Dynamic Graphs

Early methods used hand-crafted signals. CM-Sketch [7] keeps approximate edge counts and flags outliers; SpotLight [8] measures distances between graph sketches. Deep methods followed: AddGraph [3] combines a GCN with a GRU, StrGNN [4] learns from enclosing subgraphs, and TADDY [5] puts spatial and temporal context into one Transformer.

GeneralDyG [9] and GADY [10] push accuracy further. A recent unified benchmark compares these families under one protocol [6]. None of these detectors explains its predictions.

B. Explainability for Graph Neural Networks

GNNExplainer [11] learns a soft mask over edges and features; PGExplainer [12] learns this mask with a shared network. GraphSVX [13] estimates Shapley values for GNNs with a "Smarter" coalition sampler and a weighted least-squares surrogate, in the spirit of KernelSHAP [14]. We extend GraphSVX to TADDY's temporal tokens, which requires a new player definition, a new perturbation and a dual aggregation.

C. Explainability for Temporal and Dynamic GNNs

T-GNNExplainer [15] searches for important past events with Monte Carlo tree search, and TempME [16] finds important temporal motifs; both target continuous-time models. DyExplainer [17] and TGIB [18] build explanations into training. Han *et al.* [19] explain anomalies in dynamic *heterogeneous* graphs through relation evolution. None of these applies to a frozen, discrete-snapshot token model such as TADDY.

III. Preliminaries

A. Dynamic Graphs and Anomaly Detection

Definition 1 (Dynamic Graph)**.** *A dynamic graph is a sequence $\mathcal{G} = \{G_1, \dots, G_T\}$ where each $G_t = (V, E_t)$ shares the node set V and has a time-varying edge set $E_t \subseteq V \times V$.*

Definition 2 (Edge-Level Anomaly Detection)**.** *A scorer $f : (e, t) \mapsto [0, 1]$ gives each edge $e = (u, v) \in E_t$ at snapshot t a score $s_{e,t}$. Edges above a threshold τ are flagged.*

B. TADDY

TADDY [5] encodes each candidate edge (u, v) as a sequence of $D = (k + 2) \times W$ tokens, where k is the number of neighbours and W the lookback window. At each lookback step $l \in \{0, \dots, W-1\}$ there are two endpoint tokens (u, v) and the k most important neighbours, ranked by personalised PageRank. Each token sums three learned embeddings:

$$\mathbf{h}_{i,l} = \mathbf{E}_{\text{pos}}[\text{pos}_{i,l}] + \mathbf{E}_{\text{hop}}[\text{hop}_{i,l}] + \mathbf{E}_{\text{time}}[\text{time}_{i,l}]. \quad (1)$$

The token matrix $\mathbf{H}_e \in \mathbb{R}^{D \times d}$ goes through an L-layer Transformer encoder, and the score is

$$s_{e,t} = \sigma\Big(\mathbf{W}_c \cdot \tfrac{1}{k+1} \textstyle\sum_{i=0}^{k} \mathbf{h}_i^{(L)}\Big). \quad (2)$$

C. Shapley Values

For players $\mathcal{N} = \{1, \dots, M\}$ and a value function $v : 2^{\mathcal{N}} \to \mathbb{R}$, the Shapley value of player j is

$$\phi_j = \sum_{S \subseteq \mathcal{N} \setminus \{j\}} \frac{|S|!\,(M-|S|-1)!}{M!} \big[v(S \cup \{j\}) - v(S)\big]. \quad (3)$$

Shapley values are the unique attribution that satisfies efficiency, symmetry, linearity and the null-player axiom [20]. GraphSVX [13] approximates them with a surrogate weighted by the Shapley kernel $w(S) = \frac{M-1}{\binom{M}{|S|}\,|S|\,(M-|S|)}$ [14].

Algorithm 1 Dual Spatial-Temporal Shapley Attribution

Require: Trained TADDY (frozen); edge e^*; snapshot t^*; budget N_s
Ensure: $\boldsymbol{\phi}_{\text{spatial}} \in \mathbb{R}^{k+2}$, $\boldsymbol{\phi}_{\text{temporal}} \in \mathbb{R}^{W}$
$s^* \leftarrow \sigma\big(\text{TADDY_logit}(e^*, t^*)\big)$
$\mathbf{H} \leftarrow \text{EdgeEncoding}(e^*, t^*) \in \mathbb{R}^{D \times d}$
Sample $\mathbf{Z} \in \{0, 1\}^{N_s \times D}$ (Smarter); kernel weights $\mathbf{w}$
for $i = 1$ **to** N_s **do**
  $\tilde{\mathbf{H}}^{(i)} \leftarrow$ Eq. (4) on $\mathbf{H}$ with mask $\mathbf{Z}[i, \cdot]$
  $f_i \leftarrow \sigma\big(\text{Transformer_logit}(\tilde{\mathbf{H}}^{(i)})\big)$
end for
Fit kernel-weighted Ridge on $(\mathbf{Z}, \mathbf{f}, \mathbf{w})$; get $\hat{\boldsymbol{\beta}} \in \mathbb{R}^{D}$
$\phi_{\text{spatial}}[r] \leftarrow \sum_{j:\,\text{role}(j)=r} \hat{\beta}_j$ for each role r
$\phi_{\text{temporal}}[l] \leftarrow \sum_{j:\,\text{step}(j)=l} \hat{\beta}_j$ for each step l
return $\boldsymbol{\phi}_{\text{spatial}}$, $\boldsymbol{\phi}_{\text{temporal}}$

IV. The Explainability Framework

A. Design Principles

Post-hoc, zero modification. The framework wraps a trained, frozen TADDY model. Weights, training and inference are untouched, so detection is preserved exactly.

Dual decomposition. Attributions are aggregated along two axes: *spatial* (which token role: u, v, or the neighbour of PageRank rank r) and *temporal* (which lookback step).

Graph-aware perturbation. Masking happens in TADDY's dense embedding space, not in its discrete index space (Section IV-B). Fig. 1 shows the pipeline.

B. Graph-Aware Coalition Masking

Index-space masking is misleading for TADDY: index 0 in the intimacy-rank table $\mathbf{E}_{\text{pos}}$ is the *highest-ranked* PageRank neighbour, not a neutral token. Setting excluded indices to zero therefore feeds the Transformer a strong neighbour signal for every excluded token and biases the coalition scores.

We instead replace inactive tokens with the mean embedding of the active coalition:

$$\tilde{\mathbf{h}}_j = \begin{cases} \mathbf{h}_j & j \in S, \\ \bar{\mathbf{h}}_S = \frac{1}{|S|} \sum_{i \in S} \mathbf{h}_i & j \notin S. \end{cases} \quad (4)$$

The masked sequence skips EdgeEncoding and enters the Transformer directly. It stays inside the convex hull of the active token embeddings, so the model sees inputs close to its training distribution. This is the natural counterpart of GraphSVX's neutral baseline for a token-based model, and the ablation in Section V-D confirms that it matters.

C. Algorithm

Algorithm 1 gives the full procedure.

D. Surrogate Fitting and Dual Decomposition

The Smarter sampler always includes the full coalition $\mathbf{1}$ and the empty coalition $\mathbf{0}$ (kernel weight capped at 10^3, which enforces efficiency in practice), then all $2D$ singleton

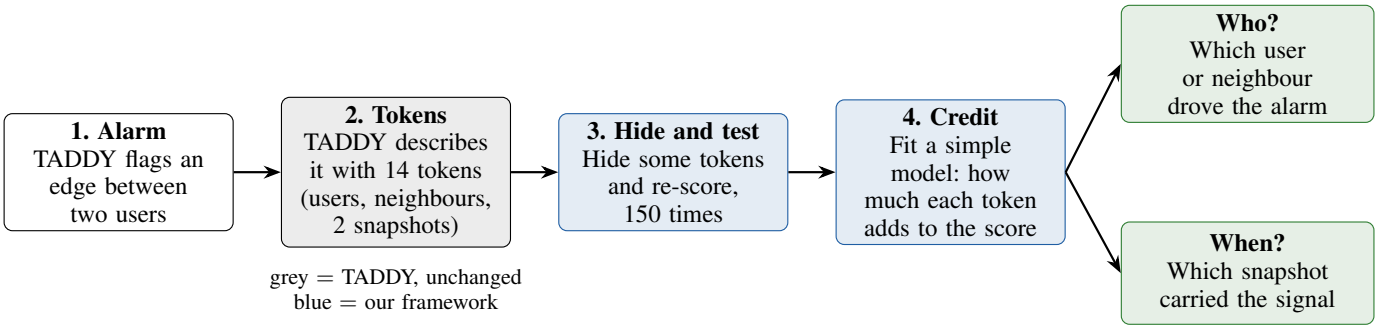


Fig. 1. How the framework works. TADDY stays unchanged. For each alarm, we hide different groups of tokens, see how the score moves, and give each token its share of the score. Adding up these shares by user role answers *who*; adding them up by snapshot answers *when*.

perturbations (one token off, one token on), and fills the rest of the budget with random coalitions. The surrogate minimises

$$\hat{g} = \arg\min_{g} \sum_{i=1}^{N_s} w_i \big(g(\mathbf{Z}_i) - f_i\big)^2 + \lambda \|\boldsymbol{\beta}\|^2, \qquad (5)$$

and its coefficients $\hat{\boldsymbol{\beta}}$ are the approximate Shapley values. Summing them by role and by step gives

$$\phi_{\text{spatial}}[r] = \sum_{j:\,\text{role}(j)=r} \hat{\beta}_j, \qquad \phi_{\text{temporal}}[l] = \sum_{j:\,\text{step}(j)=l} \hat{\beta}_j, \qquad (6)$$

where $\text{role}(j) \in \{u, v, n_1, \ldots, n_k\}$ and n_r is the neighbour of PageRank rank r. Endpoint tokens usually receive larger values because TADDY pools the first $k{+}1$ tokens in Eq. (2). The neighbour values measure what each counterparty adds beyond the edge itself, which is the signal an analyst needs for triage.

E. Explanation Quality Metrics

All metrics need an attribution vector; for TADDY alone they are undefined. Let s_{top_k} be the score with only the k highest-$|\hat{\beta}|$ tokens active, and $s_{\overline{\text{top}_k}}$ the score with all tokens *except* them, both computed with Eq. (4). Following the rationale literature [21]: **Fidelity** $\text{Fid} = 1 - |s^* - s_{\text{top}_k}|$; **Sufficiency** $\text{Suff} = s_{\text{top}_k}$ (are the top-k tokens enough to keep the alarm?); **Comprehensiveness** $\text{Comp} = s^* - s_{\overline{\text{top}_k}}$ (how much the score falls when they are removed; positive means they are necessary); **Sparsity** $1 - k/D$. All metrics are in probability space.

V. Experiments

A. Datasets and Setup

We use three dynamic social graph benchmarks from the TADDY paper [5] (Table I). Digg (30,360 nodes) was left out because the spectral clustering used for anomaly injection exceeded our memory budget.

TADDY. k=5, W=2, d=32, L=2 layers, 2 heads, learning rate 10^{-3}, weight decay $5{\times}10^{-4}$, 100 epochs, Adam, 50/50 train/test snapshot split, seed 1, all as in [5]. No tuning was done for the explainability study. The Transformer was re-implemented in pure PyTorch for Python 3.12, with the same architecture and training.

TABLE I
Datasets. Anomalies are synthetic cross-community edges injected at 10% following TADDY's protocol [5].

Dataset	Nodes	Edges	Snapshots	Anom. rate
UCI-Message	1,899	59,835	15	10%
Bitcoin-Alpha	3,783	24,186	15	10%
Bitcoin-OTC	5,881	35,592	11	10%

Framework. N_s=150 coalitions, top-k=5, D=14, Ridge λ=0.01; sparsity is fixed at $1 - 5/14 = 0.643$.

Baselines. *Random* (Gaussian scores); *Attention*: last-layer attention received per token, averaged over heads [22]; *Gradient*: L_2 norm of the gradient of s^* with respect to each token embedding [23]; *Gradient×Input*: sum over dimensions of gradient times embedding [24]. All methods are scored with the same metrics and the same masking (Section IV-E).

Statistics. We report 95% bootstrap confidence intervals (2,000 resamples) and paired two-sided Wilcoxon signed-rank tests [25] between our method and each alternative, pooled over the 150 explained edges. Code is publicly available.[1]

B. Detection Performance

Our re-implementation stays within 0.009–0.017 AUC of the published numbers (Table III). Table IV is the starting point of this work: before it, no explanation metric can be computed for TADDY.

C. Comparison with Baselines

Table II compares all methods on the *same* trained models and the same top-50 true positives of each dataset. Fig. 2 shows UCI-Message with confidence intervals.

Three results stand out.

(i) Comprehensiveness. Our method has the highest Comprehensiveness on *all three* datasets, and the gap is significant against every baseline ($p \le 2.3{\times}10^{-11}$). Removing the tokens it selects lowers the score more than removing the tokens chosen by attention or gradients, so its top tokens are the most *necessary* ones.

[1] https://github.com/Iyad-Assaad-Nekka/Explainable-Taddy

TABLE II

CONTROLLED COMPARISON: EVERY METHOD EXPLAINS THE SAME 50 HIGHEST-SCORED TRUE POSITIVES PER DATASET ON THE SAME TRAINED TADDY MODELS (k=5, D=14). MEAN OVER EDGES; "AVG" IS THE MEAN OF THE THREE DATASETS. †: OUR METHOD IS BETTER WITH $p < 0.01$ (PAIRED WILCOXON, 150 EDGES); N.S.: $p \geq 0.05$. BECAUSE THIS IS A SEPARATE TRAINING RUN OF THE SAME CONFIGURATION, ABSOLUTE VALUES DIFFER SLIGHTLY FROM TABLES IV, V AND VI; THE COMPARISON *within* THIS TABLE IS WHAT MATTERS.

	Fidelity ↑				**Sufficiency ↑**				**Comprehensiveness ↑**			
Method	UCI	BTC-α	OTC	Avg	UCI	BTC-α	OTC	Avg	UCI	BTC-α	OTC	Avg
Random	0.8711	**0.9973**	0.9983	0.9555 (n.s.)	0.8348	0.9916	0.9946	0.9403	0.0447	−0.0002	0.0002	0.0149†
Attention [22]	0.9002	0.9959	0.9978	0.9647†	0.8459	**0.9953**	0.9968	0.9460	0.1542	0.0006	0.0005	0.0518†
Gradient [23]	0.9157	0.9968	0.9984	0.9703†	**0.9709**	0.9943	0.9968	**0.9873**	0.1530	−0.0010	0.0002	0.0507†
Grad×Input [24]	0.8704	0.9969	**0.9995**	0.9556†	0.8219	0.9943	0.9949	0.9370	0.1122	0.0002	0.0018	0.0381†
Ours	**0.9342**	0.9967	0.9980	**0.9763**	0.9624	0.9945	**0.9971**	0.9847	**0.1624**	**0.0169**	**0.0085**	**0.0626**
Ablation of our method												
Index-space masking	0.9033	0.9968	0.9987	0.9663 (n.s.)	0.8448	0.9944	0.9941	0.9444	0.1053	0.0040	0.0016	0.0370†
Random sampler	0.9253	0.9967	0.9984	0.9735 (n.s.)	0.9676	0.9944	0.9967	0.9862	0.1814	−0.0009	0.0046	0.0617†
OLS (no kernel weights)	0.9285	0.9968	0.9979	0.9744 (n.s.)	0.9746	0.9944	0.9972	0.9887	0.1927	0.0006	0.0060	0.0665 (n.s.)

TABLE III

AUC-ROC. THE EXPLAINABILITY LAYER LEAVES DETECTION UNCHANGED (ΔAUC $= 0$ BY CONSTRUCTION).

Method	**UCI**	**BTC-α**	**BTC-OTC**
TADDY [5] (published)	0.8370	0.9423	0.9425
TADDY (our implementation)	0.8284	0.9554	0.9596
+ Explainability layer (ours)	**0.8284**	**0.9554**	**0.9596**

TABLE IV

EXPLAINABILITY INTRODUCED. N/A = UNDEFINED: TADDY PRODUCES NO ATTRIBUTION, SO NO METRIC CAN BE COMPUTED. MEAN OVER THE TOP-50 TRUE POSITIVES OF EACH DATASET, AVERAGED ACROSS THE THREE DATASETS.

Capability / Metric	**TADDY**	**+ Expl. (ours)**	**Status**
Fidelity	N/A	**0.9562**	Introduced
Sufficiency	N/A	**0.9658**	Introduced
Comprehensiveness	N/A	**0.0576**	Introduced
Spatial attribution	N/A	✓	Introduced
Temporal attribution	N/A	✓	Introduced
Coalitions evaluated per edge	0	150	—

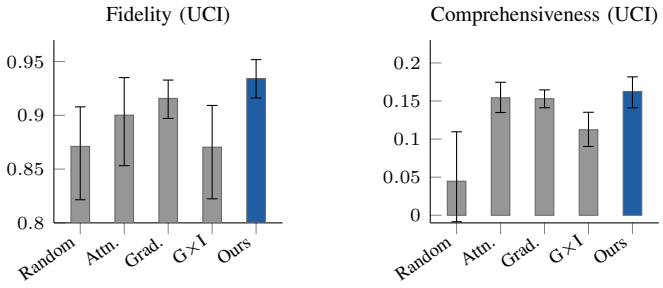


Fig. 2. UCI-Message, the dataset where the metrics are not saturated. Bars: mean over the top-50 true positives; whiskers: 95% bootstrap interval. Our method (blue) has the highest Fidelity and Comprehensiveness.

(ii) Fidelity. Our method has the best average Fidelity and is significantly better than Attention ($p = 3.7 \times 10^{-5}$), Gradient ($p = 6.1 \times 10^{-3}$) and Gradient×Input ($p = 3.9 \times 10^{-3}$). The difference with Random is not significant ($p = 0.25$), and the reason is informative: on both Bitcoin datasets *every* method, including Random, reaches Fidelity ≈ 0.997. There the top-alarm edges are scored near 1 from almost any subset of tokens, so Fidelity cannot separate methods. On UCI, where it can, our method leads (0.934 vs. 0.916 for the best baseline).

(iii) Sufficiency. Gradient is slightly higher on average (0.987 vs. 0.985), mostly on UCI. Sufficiency rewards tokens that *keep* the score high; Comprehensiveness rewards tokens whose removal *lowers* it. Our method gives the best balance of the two.

D. Ablation

The lower block of Table II removes one component at a time.

Embedding-space masking is essential. With index-space masking (excluded indices set to 0), Comprehensiveness drops by 41% (0.063 → 0.037, $p = 8.8 \times 10^{-17}$), and Sufficiency on UCI drops from 0.962 to 0.845. This confirms the analysis of Section IV-B: index 0 is not neutral.

The sampler and the kernel weights matter much less. A plain random sampler gives similar Fidelity and slightly lower Comprehensiveness ($p = 3.2 \times 10^{-3}$), mainly because it turns negative on Bitcoin-Alpha. Dropping the kernel weights (OLS) gives statistically indistinguishable results (Comprehensiveness $p = 0.67$). These parts are therefore replaceable; we keep them because they follow the standard Shapley estimator [13], [14] and preserve its efficiency anchor.

E. Expanded Evaluation Across Edge Categories

Table V evaluates four edge categories. The groups overlap: "Top-50 TP" and "Low-Conf TP" are subsets of "All TP", so the evaluation covers 2,471 distinct true positives and 150 false positives, i.e. 2,621 distinct edges.

(i) Beyond the top-50. Fidelity on all true positives stays high (0.902–0.974), so the results do not depend on picking the most confident alarms.

(ii) False positives. Fidelity on false alarms is 0.843 on UCI and $\geq$ 0.996 on the Bitcoin datasets. The framework can

TABLE V
FIDELITY ACROSS EDGE CATEGORIES. FP = FALSE POSITIVES (HIGHEST-SCORED NORMAL EDGES); LOW-CONF TP = TRUE ANOMALIES WITH THE LOWEST SCORES. TOP-50 TP AND LOW-CONF TP ARE SUBSETS OF ALL TP: 2,471 DISTINCT TP + 150 FP = 2,621 DISTINCT EDGES.

Dataset	Category	n	Fidelity	±Std
UCI	Top-50 TP	50	0.8743	0.222
UCI	All TP	691	0.9023	0.159
UCI	FP	50	0.8428	0.281
UCI	Low-Conf TP	50	0.8101	0.352
BTC-α	Top-50 TP	50	0.9957	0.001
BTC-α	All TP	706	0.9598	0.114
BTC-α	FP	50	0.9957	0.001
BTC-α	Low-Conf TP	50	0.9893	0.031
BTC-OTC	Top-50 TP	50	0.9982	0.001
BTC-OTC	All TP	1074	0.9735	0.095
BTC-OTC	FP	50	0.9977	0.001
BTC-OTC	Low-Conf TP	50	0.8158	0.300
Distinct edges		**2,621**		

TABLE VI
FIDELITY OF OUR METHOD FOR $k \in \{3, 5, 7, 10\}$ (TOP-50 TP). IT RISES WITH k ON UCI; IT IS NEAR SATURATION ON THE BITCOIN DATASETS.

Dataset	k=3	k=5	k=7	k=10
UCI-Message	0.8578	0.8720	0.8976	0.9454
Bitcoin-Alpha	0.9951	0.9957	0.9961	0.9975
Bitcoin-OTC	0.9994	0.9982	0.9981	0.9988

therefore explain *why* TADDY flagged a normal edge, which is what an analyst needs to clear a false alarm quickly.

(iii) Borderline detections. Fidelity on low-confidence true positives is 0.810–0.989, lower and more variable than on confident ones, as expected near the decision boundary.

(iv) Comprehensiveness by category. The value 0.0576 in Table IV is the mean over the top-50 true positives of the three datasets. In the controlled run (Table II) it is positive on every dataset (0.162, 0.017, 0.009). The one negative value we observed is for low-confidence true positives on Bitcoin-Alpha (-0.3995), discussed in Section VI-D.

F. Sensitivity to the Top-k Budget

Table VI shows that Fidelity on UCI rises steadily with k, which is what a correct token ranking should produce: adding the next-best tokens brings the score closer to the full-context score.

G. Worked Example

Fig. 3 shows the explanation of the highest-scored true anomaly on UCI-Message ($s^* = 0.9965$).

An analyst reads it as follows. *Spatially*, both endpoints push the score up ($+0.161$, $+0.165$), while the top-ranked neighbour n_1 pulls it down strongly (-0.293): this neighbour looks like normal context, so the alarm is about the pair (u, v) itself, not about a suspicious counterparty around it. *Temporally*, the current snapshot raises the score ($+0.074$) and the previous one lowers it (-0.075): the interaction does not

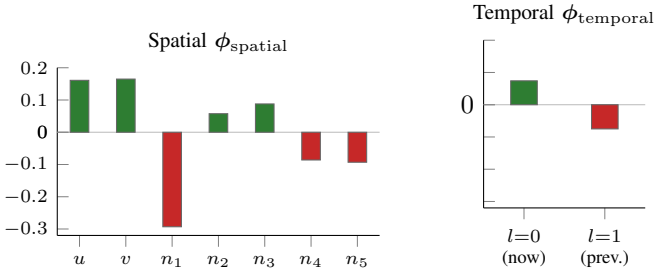


Fig. 3. Explanation of the top-scored true anomaly on UCI-Message ($s^* = 0.9965$). Green raises the anomaly score, red lowers it. Spatial values sum each role over both lookback steps; n_r is the neighbour of PageRank rank r.

fit the pair's recent history, so the anomaly appeared *now*. A scalar score gives neither answer. For the top alarms on the two Bitcoin datasets, the temporal vector also favours the current snapshot ($+0.057$ vs. $+0.043$ on Bitcoin-Alpha; $+0.018$ vs. -0.012 on Bitcoin-OTC).

H. Computational Cost

On a single CPU core (Google Colab), one explanation takes 0.18 s (0.17–0.18 s across datasets), against 0.035 ms per edge for batched TADDY inference. Gradient×Input costs 0.004 s, and attention and gradient norms are almost free. The cost of our method grows linearly with N_s, since each coalition is one Transformer forward pass, and the N_s passes can be batched on a GPU. This fits offline triage of the alarms an analyst actually reviews, not scoring every edge in a stream.

VI. DISCUSSION

A. Why the Dual Decomposition Matters

Any per-token score can be summed into spatial and temporal vectors. What our framework adds is that the summed values are coalition-based marginal contributions that satisfy the Shapley axioms (efficiency, symmetry, linearity, null player) [20], while gradients measure first-order sensitivity and attention reflects internal focus rather than effect on the output [22]. Table II shows this difference in practice: the tokens our method selects are the most necessary ones (highest Comprehensiveness on every dataset).

B. Why Post-Hoc?

We chose a post-hoc design on purpose. **Deployment:** practitioners often have trained detectors they cannot retrain; intrinsic methods such as TGIB [18], SIGNET [26] and DyExplainer [17] change the training objective, which can change detection. Our framework keeps $\Delta\text{AUC} = 0$. **Separation of concerns:** detection and explanation can be improved independently. **Reuse:** the same procedure applies to any detector with a similar token input.

C. Applicability Beyond TADDY

The framework needs only a token-based input and a forward pass from token embeddings. It applies directly to other windowed Transformer detectors that encode an edge

as a token sequence, and the dual aggregation of Eq. (6) only needs to know the role and time step of each player. For models without explicit tokens, such as recurrent GNN detectors [3] or subgraph-based detectors [4], the players must be redefined (for example as nodes of the enclosing subgraph per snapshot) and the neutral replacement must be chosen for that input space; this is the main step needed to transfer the method. In social network operations, the output answers the two questions moderators ask about an alert: *who* drove it, and *when*.

D. Borderline Detections and Negative Comprehensiveness

For low-confidence true positives on Bitcoin-Alpha, Comprehensiveness is negative (-0.3995): removing the top-k tokens *raises* the score. For these borderline edges, the tokens with the largest $|\hat{\beta}|$ are those that pull the score down, and the anomaly signal is spread across many tokens rather than concentrated in a few. Such edges are better described by the whole attribution vector than by a top-k summary, and an analyst should read them that way.

VII. Conclusion

We introduced the first post-hoc explainability framework for TADDY-style dynamic graph anomaly detectors. For each flagged edge it returns a spatial vector (which roles drove the alarm) and a temporal vector (which snapshot), with no change to detection. On the same trained models, it gives the most necessary explanations of all baselines (highest Comprehensiveness on every dataset, $p < 10^{-10}$) and the best Fidelity where the metric is not saturated. The ablation shows that its key ingredient is embedding-space masking, while the sampler and kernel weights are replaceable. An evaluation over $2{,}621$ edges and a worked example show how the two vectors support alarm triage, including false alarms. The explainability gap is not unique to TADDY: it is the default state of deep dynamic graph detectors, and the recipe used here (frozen model, meaningful players, neutral replacement in the model's own input space) offers a template for closing it.

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